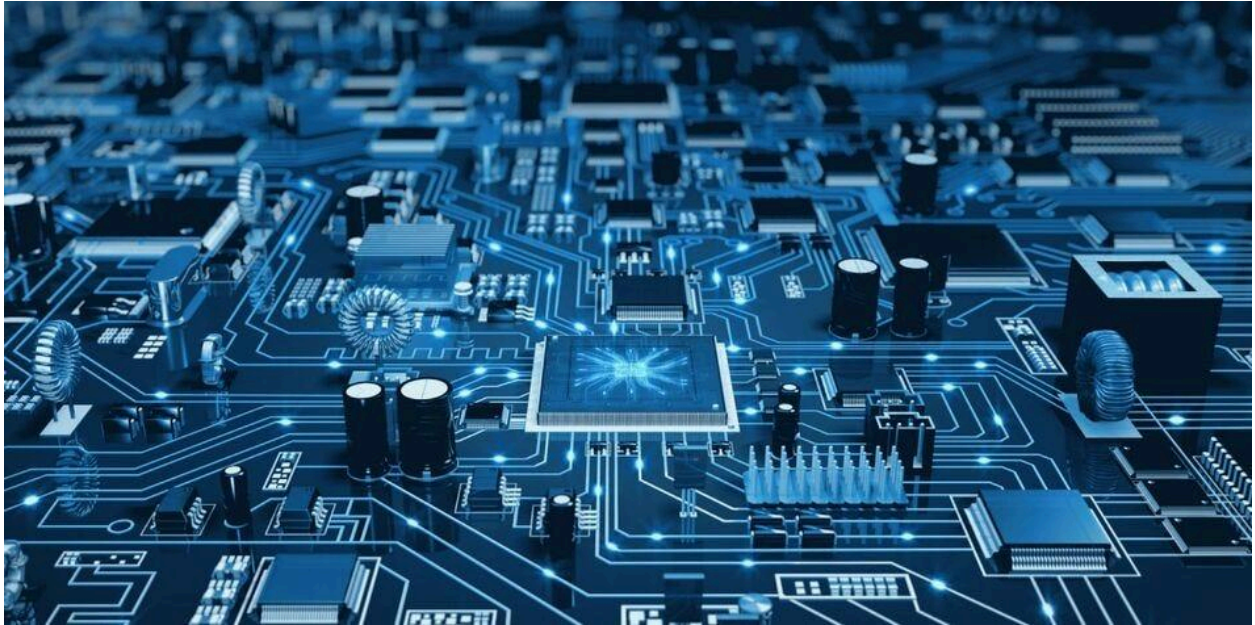


Embedded Systems and IoT project

Joint-Track Knee Rehab Sleeve

A smart compression sleeve for tracking ROM during physiotherapy



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Abstract

The Joint-Track Knee Rehab Sleeve is a smart wearable rehabilitation device designed to assist patients recovering from ACL injuries and knee surgeries. Many patients perform physiotherapy exercises incorrectly at home, resulting in delayed recovery due to improper joint angles and inconsistent repetitions. This project proposes a smart compression sleeve embedded with motion and biomechanical sensors to monitor the knee's Range of Motion (ROM) in real time.

The system calculates knee flexion angle using multiple IMUs and flex sensors, provides haptic and audio feedback when the target angle (e.g., 90°) is reached, and logs exercise data (repetitions, sets, hold duration) to a mobile application via Bluetooth. This enables doctors and physiotherapists to remotely monitor patient progress and ensure accurate rehabilitation.

Problem Statement

Patients recovering from **Anterior Cruciate Ligament Reconstruction (ACL reconstruction)**, knee replacement surgeries, or other knee injuries are usually prescribed physiotherapy exercises to restore joint mobility, improve muscle strength, and regain normal movement. While initial rehabilitation may be supervised by physiotherapists, a large portion of recovery happens at home without direct medical supervision. During these unsupervised sessions, patients often struggle to perform exercises correctly because they cannot accurately judge the required knee bending angles such as **45°, 60°, or 90°**. As a result, exercises may be performed incorrectly, which can slow down recovery or place additional strain on the healing joint.

Some common challenges faced during home-based physiotherapy include:

- Patients **over-bending or under-bending** their knees beyond the prescribed angle.
- Lack of **real-time feedback** to confirm whether the exercise is performed correctly.
- Difficulty in **tracking repetitions, sets, and hold duration** during exercises.

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- Limited ability for **doctors or physiotherapists to monitor rehabilitation progress remotely.**

To address these challenges, there is a need for a **smart, low-cost, wearable rehabilitation monitoring system** that can measure knee joint movement accurately and provide instant feedback to the user. The proposed **Joint-Track Knee Rehab Sleeve** uses multiple sensors, embedded processing, and wireless communication to monitor knee angles in real time, guide patients during exercises, and record rehabilitation data for medical review. This system aims to improve the effectiveness of home-based physiotherapy and support remote monitoring by healthcare professionals.

Board Used

The primary board used will be - ESP32 Dev Module. It collects sensor data, performs angle calculation and transmits data to the mobile app. The specification of the board are as follows:

Microcontroller	Tensilica Xtensa Dual-Core 32-bit
Clock Speed	240 MHz
Operating Voltage	3.3 V
Bluetooth	BLE + Classic
ADC Pins	18 channels

The reasons for the selection of ESP32 Dev board are:

- It includes built-in Bluetooth and WiFi connectivity, enabling seamless wireless communication
- It features a dual-core processor capable of operating up to 240 MHz, which allows efficient real-time processing of multiple sensor inputs simultaneously.
- The board is designed for low power consumption, making it highly suitable for battery-powered wearable devices used in continuous health monitoring.

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- The board provides multiple GPIO pins along with ADC, I2C, SPI, and UART interfaces, allowing easy integration of several sensors and actuators.
 - The ESP32 development module is compact, lightweight, and cost-effective, making it ideal for integration into small wearable rehabilitation systems.

Sensors Used

1. IMU Sensor (Upper Leg):

- 1.1. Sensor Model : MPU6050
- 1.2. Sensor Name : 6-Axis Inertial Measurement Unit
- 1.3. Usage :
 - 1.3.1. It measures the orientation of the thigh for the upper leg.
 - 1.3.2. It provides the angular velocity and acceleration data.
 - 1.3.3. It helps to calculate the thigh angle in real time.

1.4 Board Wiring :

<u>MPU6050 Pin</u>	<u>ESP32 Pin</u>
VCC	3.3 V
GND	GND
SDA	GPIO 21
SCL	GPIO 22
AD0	GND

2. IMU Sensor (Lower Leg):

- 2.1. Sensor Model : MPU6050
- 2.2. Sensor Name : 6-Axis Inertial Measurement Unit
- 2.3. Usage :
 - 2.3.1. It measures the orientation of the shin (lower leg).
 - 2.3.2. It helps in finding the knee angle ; Knee angle = IMU angle - Lower IMU angle

2.4. Board Wiring :

<u>MPU6050 Pin</u>	<u>ESP32 Pin</u>
VCC	3.3 V
GND	GND
SDA	GPIO 21
SCL	GPIO 22
AD0	3.3 V

3. **Flex Sensor:**

3.1. Sensor Model : Spectra Symbol 4.5" Flex Sensor

3.2. Sensor Name : Resistive Bend Sensor

3.3. Usage :

3.3.1. It measures the direct bending of the knee.

3.3.2. It will act as a validation for IMU-based angles.

3.3.3. It provides analog voltage proportional to bend.

3.4. Board Wiring :

<u>Flex Sensor</u>	<u>ESP32 Pin</u>
One end	3.3 V
Other end	GPIO 34 (ADC)
10,000 Ohm resistor	Between GPIO 34 & GND

4. **Force Sensor:**

4.1. Sensor Model : FSR402

4.2. Sensor Name : Force Sensitive Resistor

4.3. Usage :

4.3.1. It detects weight-bearing during exercises.

4.3.2. It ensures correct pressure while squatting or standing.

4.3.3. Prevents improper loading of the knee.

4.4. Board Wiring :

<u>FSR Pin</u>	<u>ESP32 Pin</u>
One end	3.3 V
Other end	GPIO 35 (ADC)
10,000 Ohm resistor	Between GPIO 35 & GND

5. Heart Rate Sensor:

5.1. Sensor Model : MAX30102

5.2. Sensor Name : Pulse Oximeter & Heart Rate Sensor

5.3. Usage :

5.3.1. It monitors heart rate during therapy.

5.3.2. It tracks patient exertion level.

5.3.3. It ensures safe rehabilitation intensity.

5.4. Board Wiring :

<u>MAX30102</u>	<u>ESP32</u>
VCC	3.3 V
GND	GND
SDA	GPIO 21
SCL	GPIO 22

6. Temperature Sensor:

6.1. Sensor Model : DS1820B

6.2. Sensor Name : Digital Temperature Sensor

6.3. Usage :

6.3.1. It measures knee surface temperature.

6.3.2. It detects inflammation or swelling.

6.3.3. It helps monitor recovery conditions.

6.4. Board Wiring :

<u>DS18B20</u>	<u>ESP32</u>
VCC	3.3 V
GND	GND
DATA	GPIO 4
4.7k Ohm resistore	Between DATA & 3.3 V

7. EMG Sensor:

- 7.1. Sensor Model : MyoWare 2.0 Muscle Sensor
- 7.2. Sensor Name : Surface Electromyography (EMG) Sensor
- 7.3. Usage :
 - 7.3.1. It measures muscle activation in quadriceps.
 - 7.3.2. It ensures correct muscle engagement.
 - 7.3.3. It detects muscle fatigue or weak contraction.
- 7.4. Board Wiring :

<u>EMG Pin</u>	<u>ESP32</u>
VCC	3.3 V
GND	GND
SIG	GPIO 33 (ADC)

Actuators Used

1. Piezoelectric Buzzer:

- 1.1. Model - KY-012 Active Buzzer Module
- 1.2. Usage - The buzzer provides **audio feedback** to notify the patient when the knee reaches the target rehabilitation angle (for example 90°). It can also alert the user if the knee is bent beyond the safe range or if the exercise is performed incorrectly.
- 1.3. ESP32 Wiring -

<u>Buzzer Pin</u>	<u>ESP32 Pin</u>
VCC	3.3 V
GND	GND
Signal	GPIO 25

2. LED indicator:

- 2.1. Model - 5mm High Brightness LED
- 2.2. Usage - LED indicators provide **quick visual feedback** for different system states, such as: Green LED for correct knee angle reached and red LED for incorrect movement or over-bending.
- 2.3. ESP32 Wiring -

<u>LED Pin</u>	<u>ESP32 Pin</u>
Anode	GPIO 27
Cathode	GND

3. Haptic Feedback Motor:

- 3.1. Model - 10mm Coin Vibration Motor (ERM Motor)
- 3.2. Usage - The vibration motor provides **haptic feedback** when the correct knee angle is reached or when the patient needs to adjust the movement. This is useful for wearable devices because the user can feel the feedback directly without needing to look at a screen.
- 3.3. ESP32 Wiring -

<u>Motor Pin</u>	<u>ESP32 Pin</u>
Motor Positive	3.3 V
Motor Negative	NPN Transistor
Transistor Base	GPIO 26
Transistor Emitter	GND

Architecture Diagram

The architecture of the **Joint-Track Knee Rehab Sleeve** consists of multiple sensors, a central processing unit, feedback actuators, and a mobile monitoring system. Various sensors including two **MPU6050** inertial measurement units (placed on the upper and lower leg), a flex sensor, **Interlink FSR402**, **MyoWare 2.0 Muscle Sensor**, **DS18B20**, and **MAX30102** collect real-time motion and physiological data from the knee region. All sensor data is transmitted to the **ESP32**, which acts as the central controller of the system. The ESP32 processes this information to calculate the knee joint angle, track repetitions and hold time during physiotherapy exercises, and determine whether the movement is performed correctly. Based on the analysis, the system activates feedback actuators such as a buzzer, vibration motor, LED indicator, and OLED display to guide the patient during rehabilitation. Additionally, the processed data is transmitted through Bluetooth Low Energy (BLE) to a mobile application, where exercise progress can be recorded and remotely monitored by doctors or physiotherapists.

